

BLAST

Basic Local Alignment Search Tool

- Blast在线
- 本地版Blast

NCBI Databases

Results found in 35 databases for: **p53**

输入文本，检索信息——Entrez

Literature

Bookshelf	2,253
MeSH	185
NLM Catalog	127
PubMed	94,494
PubMed Central	198,851

Genes

Gene	14,861
GEO DataSets	17,841
GEO Profiles	2,433,072
HomoloGene	38
PopSet	152
UniGene	414

Genetics

ClinVar	1,986
dbGaP	0
dbSNP	6,148
dbVar	4,409
GTR	463
MedGen	138
OMIM	677

Proteins

Conserved Domains	257
Identical Protein Groups	9,470
Protein	94,815
Protein Clusters	8
Sparcle	1,003
Structure	1,319

Genomes

Assembly	1
BioCollections	0
BioProject	1,501
BioSample	4,655
Genome	7
Nucleotide	87,994
Probe	3,712
SRA	7,042
Taxonomy	19

Chemicals

BioSystems	5,847
PubChem BioAssay	7,834
PubChem Compound	16
PubChem Substance	1,557

生物信息的获取方式

- 1、输入文本，检索信息——Entrez
- 2、输入序列，检索信息——BLAST

Web Access

Bioinformatics database

Sequence



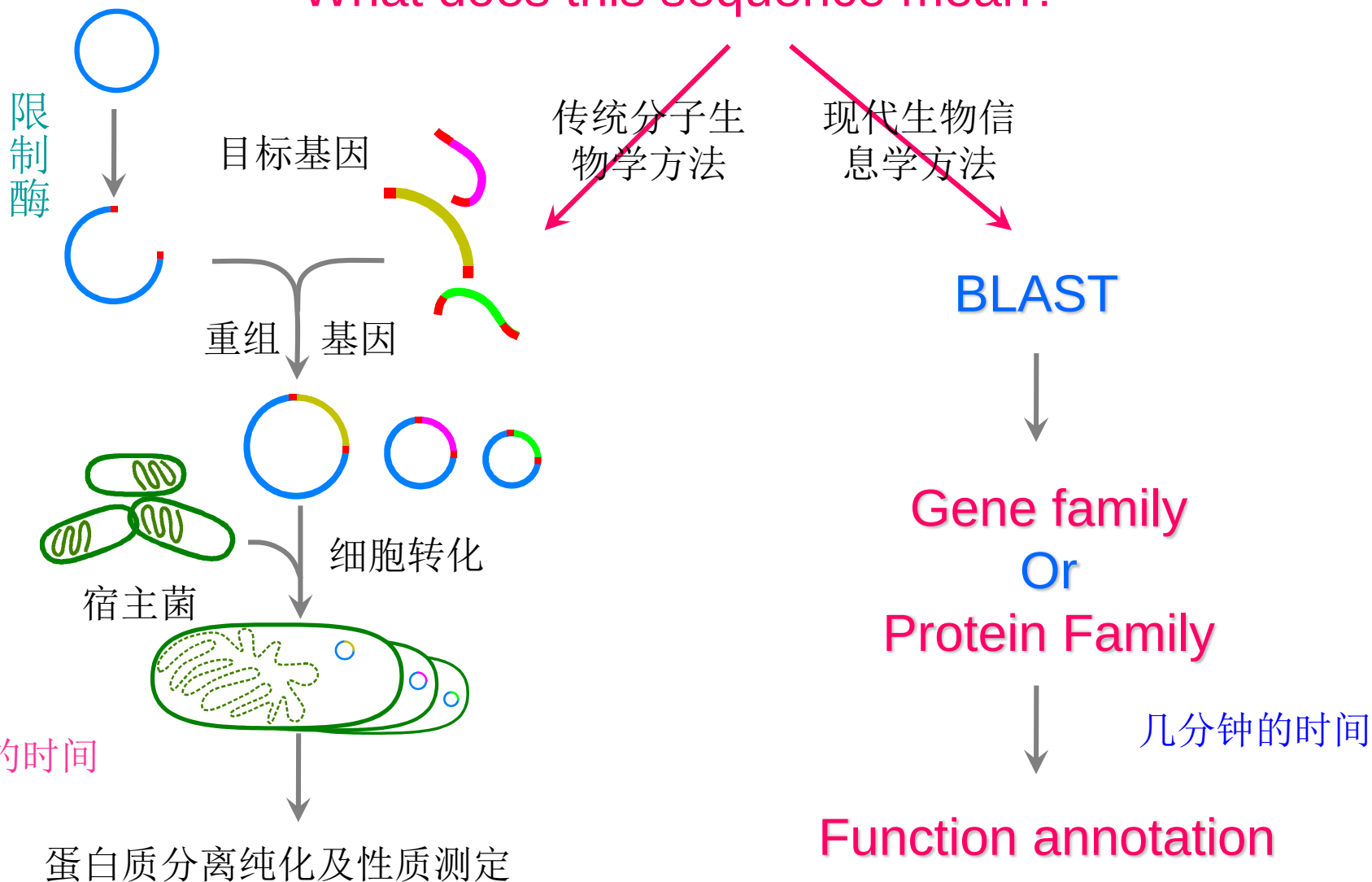
```
1 ttgctctgtg tatcgtaaat ttaagattta atgtatttat tgaagaatta aaagatatac
61 tgcaaaaaga gagagaatag aaggaatctc tttatatact gtaagtaaag gagataagag
121 agctagaagc ggaagctaata agctcttttt cgctaataaa ttatataata aagatagaac
181 gagtgaatg taggctagat tggatgagag aaaggaaaag caagaatgga aaaacagtca
241 tggatggagc tattgccgat tgatcgttat aaagtaagtg cgaaagggtt gttacataat
301 tacgaccgga aagtattaac gatgttgtat cagccgttaa taggtagtag agcttttagc
361 ttatacatga cactatgggg agagttagag caagatcgtg tgtttggaaa agagaataca
```

```

1  ttgctctgtg  tatcgtcaat  ttaagattta  atgtatttat  tgaagaatta  aaagatatac
61  tgcaaaaaga  gagagaatag  aaggaatctc  tttatatact  gtaagtaaac  gagataagag
121  agctagaagc  ggaagctaata  agctcttttt  cgctaataaa  ttatataata  aagatagaac
181  gagtgaatg  taggctagat  tggatgagag  aaaggaaaag  caagaatgga  aaaacagtca
241  tggatggagc  tattgccgat  tgatcgttat  aaagtaagtg  cgaaagggtt  gttacataat
301  tacgaccgga  aagtattaac  gatgttgat  cagccgttaa  taggtagtag  agcttttagc
361  ttatacatga  cactatgggg  agagttagag  caagatcgtg  tgtttggaag  agagaataca

```

What does this sequence mean?



www.ncbi.nlm.nih.gov

NCBI Resources ▾ How To ▾



National Center for
Biotechnology Information

Search All Databases ▾

Search

Clear

NCBI Home

Site Map (A-Z)

All Resources

Chemicals & Bioassays

Data & Software

DNA & RNA

Domains & Structures

Genes & Expression

Genetics & Medicine

Genomes & Maps

Homology

Literature

Proteins

Sequence Analysis

Taxonomy

Training & Tutorials

Variation

Welcome to NCBI

The National Center
for Biotechnology Information
health by providing:

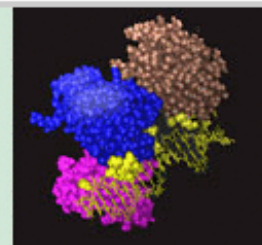
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Get Started

- [Tools](#): Analyze data using NCBI software
- [Downloads](#): Get NCBI data or software
- [How-To's](#): Learn how to accomplish specific tasks at NCBI
- [Submissions](#): Submit data to GenBank or other NCBI databases

3D Structures

Explore three-dimensional structures of proteins, DNA, and RNA molecules. Examine sequence-structure relationships, active sites, molecular interactions, biological activities of bound chemicals, and associated biosystems.



1 2 3 4

BLAST

Popular Resources

- [BLAST](#)
- [Bookshelf](#)
- [Gene](#)
- [Genome](#)
- [Nucleotide](#)
- [OMIM](#)
- [Protein](#)
- [PubChem](#)
- [PubMed](#)
- [PubMed Central](#)
- [SNP](#)

NCBI News

[NCBI's YouTube Channel has "How To" Videos](#)

15 Nov 2010

A growing set of videos can be found at YouTube, including

[NCBI Workshop: A Practical Guide to Genome-Scale Data](#)

03 Nov 2010

Presented at the ASHG Meeting

blast.ncbi.nlm.nih.gov

► NCBI/ BLAST Home

BLAST finds regions of similarity between biological sequences. [more...](#)

[Learn more](#) about how to use the new BLAST design

BLAST Assembled Genomes

Choose a species genome to search, or [list all genomic BLAST databases](#).

- | | | |
|--------------------------------------|---|---------------------------------|
| Human | Oryza sativa | Gallus gallus |
| Mouse | Bos taurus | Pan troglodytes |
| Rat | Danio rerio | Microbes |
| Arabidopsis thaliana | Drosophila melanogaster | Apis mellifera |

Basic BLAST

Choose a BLAST program to run.

[nucleotide blast](#)

Search a **nucleotide** database using a **nucleotide** query
Algorithms: blastn, megablast, discontinuous megablast

[protein blast](#)

Search **protein** database using a **protein** query
Algorithms: blastp, psi-blast, phi-blast

[blastx](#)

Search **protein** database using a **translated nucleotide** query

News

BLAST 2.2.18 now available

The binaries for BLAST 2.2.18 are now available at our [FTP site](#)

2008-04-01 07:50:00

[More BLAST news...](#)

Tip of the Day

How to Search Custom Databases in Web-Blast Using Entrez Queries

A powerful feature of the BLAST Web interface is the ability to limit BLAST searches to a subset of any database using a standard Entrez query. Skillful use of Entrez queries allows the equivalent of on-the-fly construction of databases of exact composition

[More tips...](#)


BLAST
Basic Local Alignment Search Tool

[Home](#)
[Recent Results](#)
[Saved Strategies](#)
[Help](#)

▶ [NCBI/ BLAST/ blastn suite](#)

[blastn](#)
[blastp](#)
[blastx](#)
[tblastn](#)
[tblastx](#)

BLASTN programs search nucleotide databases using a nucleotide query. [more...](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) ?

Clear

Query subrange ?

From

To

Or, upload file

浏览...

Job Title

Enter a descriptive title for your BLAST search ?

☐ Align two or more sequences ?

Choose Search Set

Database

☒ Human genomic + transcript
 ☐ Mouse genomic + transcript
 ☐ Others (nr etc.):

Human genomic plus transcript (Human G+T) ▼ ?

Exclude Optional

☐ Models (XM/XP)
 ☐ Uncultured/environmental sample sequences

Entrez Query Optional

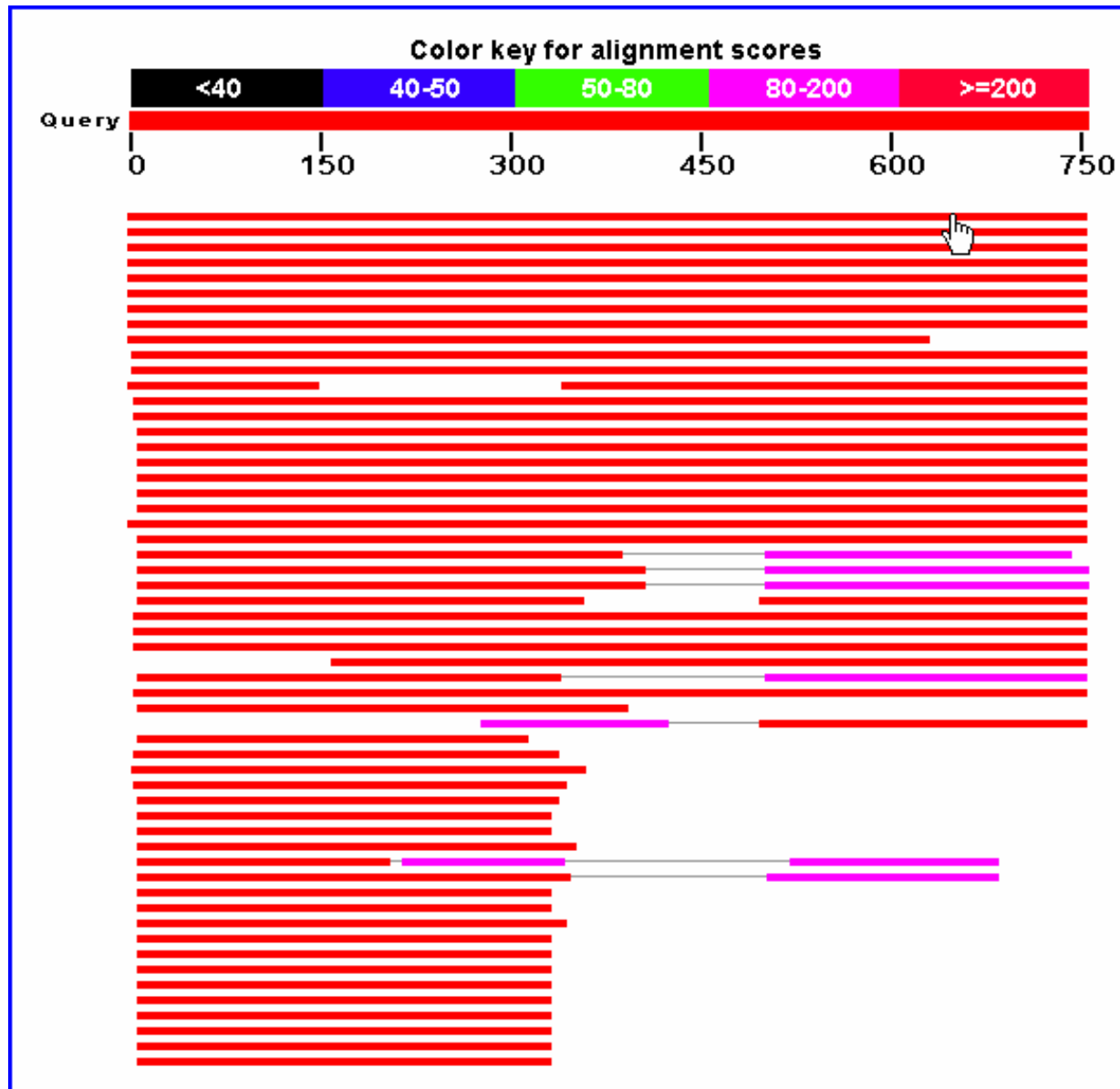
Enter an Entrez query to limit search ?

Program Selection

ENTER
Sequences
Here

BLAST

Basic Local Alignment Search Tool



- ◆ Sequence similarity
- ◆ BLAST algorithm
- ◆ BLAST statistics
- ◆ BLAST output

Sequence Similarity Searching

- To identify and annotate sequences
- To evaluate evolutionary relationships
- Other:
 - model genomic structure (e.g., Spidey)
 - check primer specificity *in silico*

BLAST: NCBI's tool

科学的方法： 可以让我们研究我们不懂的数据！ ——比较的方法

全局比对和局部比对算法的差异

全局比对

Needleman-Wunsch算法;

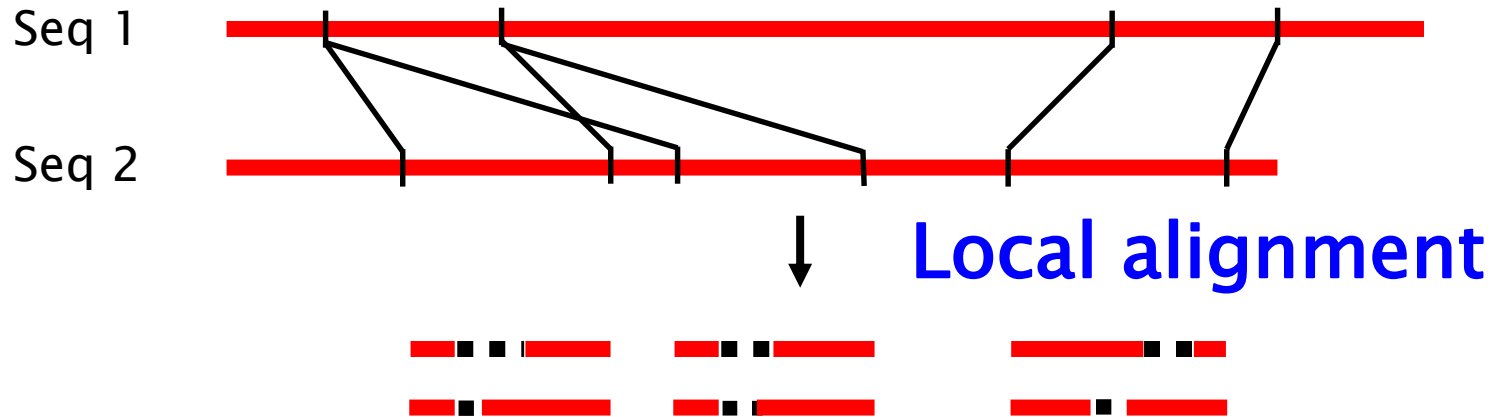
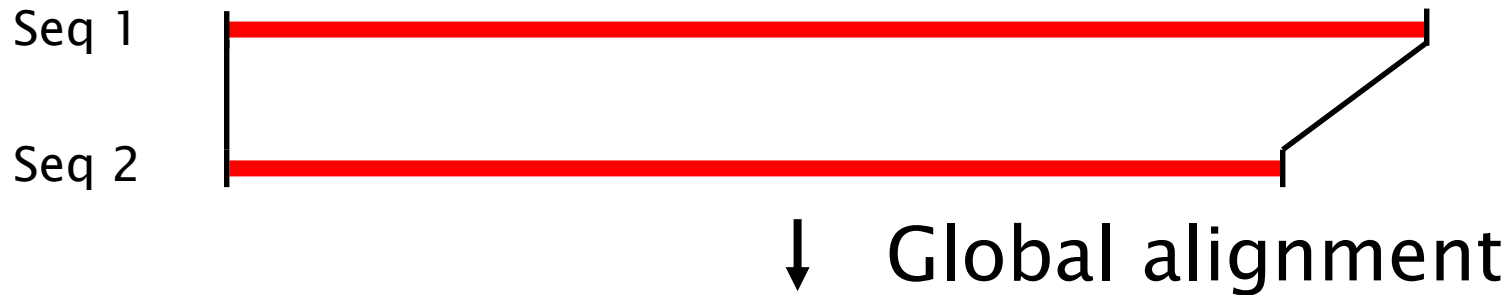
$$S_{ij} = \max \begin{cases} S_{i-1, j-1} + \sigma(x_i, y_j) \\ S_{i-1, j} + d \text{ (从上到下)} \\ S_{i, j-1} + d \text{ (从左到右)} \end{cases}$$

局部比对

Smith-Waterman算法;

$$S_{ij} = \max \begin{cases} S_{i-1, j-1} + \sigma(x_i, y_j) \\ S_{i-1, j} + d \text{ (从上到下)} \\ S_{i, j-1} + d \text{ (从左到右)} \\ 0 \end{cases}$$

Global vs Local Alignment



检索海量数据



Database

Posted date	Apr 19, 2016 8:29 AM
Number of letters	31,346,824,377
Number of sequences	85,563,261

How BLAST Works

Heuristic approach

1. Make lookup table of “words” for query
2. Scan database for hits
3. Ungapped extensions of hits
4. Gapped extensions

Nucleotide Words

Query: **GTACTGGACATGGACCCTACAGGAA**

11-mer

GTACTGGACAT

TACTGGACATG

ACTGGACATGG

CTGGACATGGA

TGGACATGGAC

GGACATGGACC

GACATGGACCC

ACATGGACCCT

. . .

Make a lookup
table of words

Protein Words

Query: **GTQITVEDLIFYNIATRRKALKN**

GTQ

Word size = 3 (default)

TQI

Word size can only be 2 or 3

QIT

ITV

TVE

VED

EDL

DLF

...

Make a lookup
table of words

Protein Words

Query: **GTQITVEDLIFYNIATRRKALKN**

GTQ

TQI

QIT

Neighborhood Words

ITV → LTV, MTV, ISV, LSV, etc.

TVE

[-f 11 = blastp default]

VED

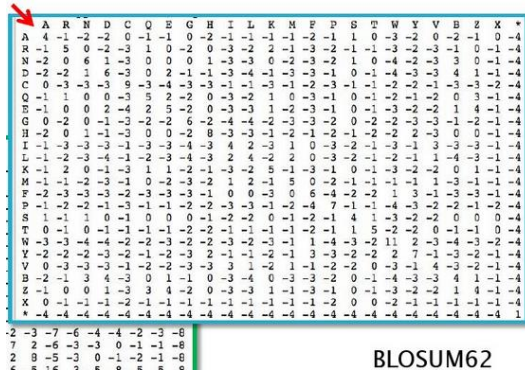
EDL

DLF

...

Make a lookup
table of words

L N K C K T P Q G Q R



BLOSUM62

临界值T=13

P Q G 7+5+6=18 单词

P E G 7+2+6=15

P R G 7+1+6=14

P K G 7+1+6=14

P N G 7+0+6=13

P M G 7+0+6=13

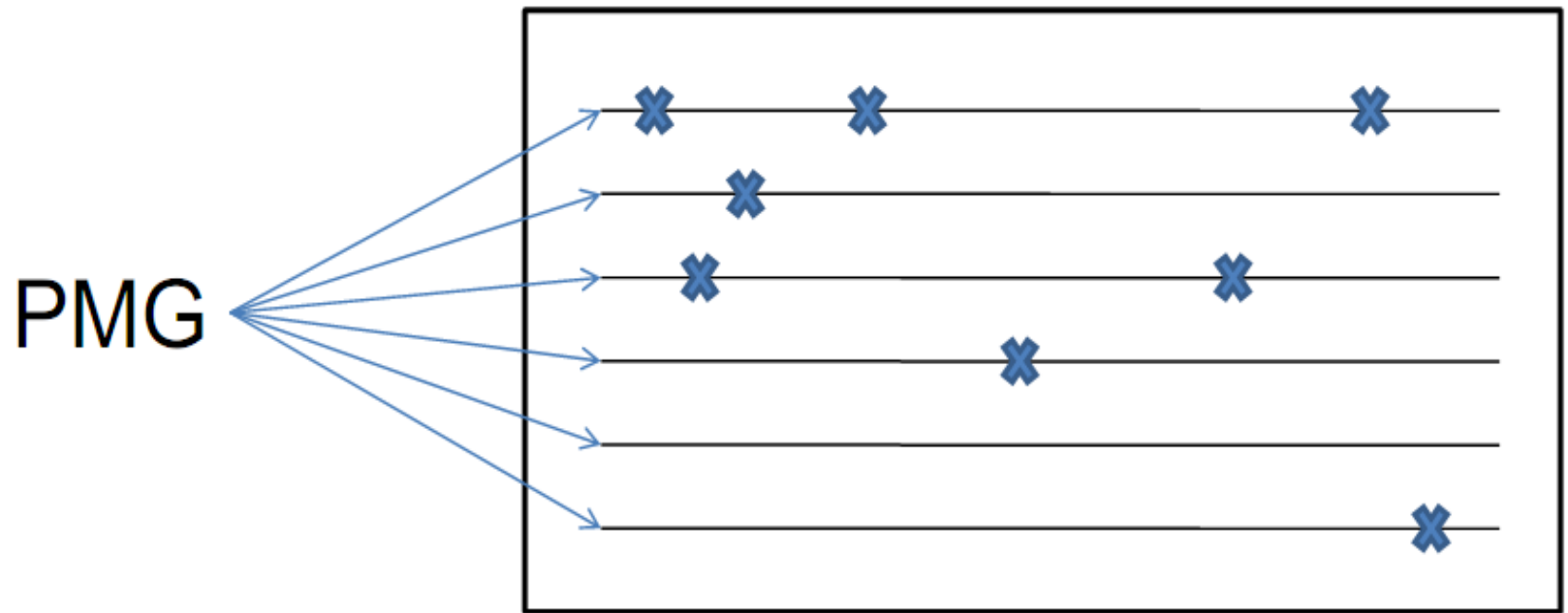
P Q A 7+5+0=12

P Q N 7+5+0=12

etc.

邻居单词

Search for Hits



Minimum Requirements for a Hit

ATCGCCATGCTTAATTGGGCTT

CATGCTTAATT

one exact match

- Nucleotide BLAST requires one exact match
- Protein BLAST requires two neighboring matches within 40 aa

GTQITVEDLFYNI

SEI

YYN

neighborhood words
[-A 40 = blastp default]

two matches

BLASTP Summary

example query words

Query: IETVYAAYLPKNTHPFL**YLS**LEISPQNVDVNVHPTKHEV**HFL**HEESILEV...

YLS	15	Neighborhood words	HFL	18	Neighborhood score threshold
YLT	12		HFV	15	
YVS	12		HFS	14	
YIT	10		HWL	13	
etc	...		NFL	13	
			DFL	12	
			HWV	10	T (-f) = 11
			etc	...	

Query 1 IETVYAAYLPKNTHPFL**YLS**LEISPQNVDVNVHPTKHEV**HFL**HEESI 47
 +E YA YL K F+**YLS**L +SP+ +DVNVHP+K **VHFL**+++ I
 Sbjct 287 LEETYAKYLHKGASYFV**YLS**LNMSPEQLDVNVHPSKRIV**HFL**YDQEI 333

High-scoring pair (HSP)

高得分配对片段

Gapped extension with trace back

有空位延伸(trace back)

Query 1 IETVYAAYLPKNTHPFL**YLS**LEISPQNVDVNVHPTKHEV**HFL**HEESI-LEV... 50
 +E YA YL K F+**YLS**L +SP+ +DVNVHP+K **VHFL**+++ I + +
 Sbjct 287 LEETYAKYLHKGASYFV**YLS**LNMSPEQLDVNVHPSKRIV**HFL**YDQEIATSI... 337

Final HSP

打分及统计学分析

PVLYVEHRS **SCR**VERRRRAEELHSAT
ATWVIEHTS **SCR**VYGRNAAALSQAI

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V	B	Z	X	*
A	4	-1	-2	-2	0	-1	-1	0	-2	-1	-1	-1	-1	-2	-1	1	0	-3	-2	0	-2	-1	0	-4
R	-1	5	0	-2	-3	1	0	-2	0	-3	-2	2	-1	-3	-2	-1	-1	-3	-2	-3	-1	0	-1	-4
N	-2	0	6	1	-3	0	0	0	1	-3	-3	0	-2	-3	-2	1	0	-4	-2	-3	3	0	-1	-4
D	-2	-2	1	6	-3	0	2	-1	-1	-3	-4	-1	-3	-3	-1	0	-1	-4	-3	4	1	-1	-4	
C	0	-3	-3	-3	9	-3	-4	-3	-3	-1	-1	-3	-1	-2	-3	-1	-1	-2	-2	-1	-3	-3	-2	-4
Q	-1	1	0	0	-3	5	2	-2	0	-3	-2	1	0	-3	-1	0	-1	-2	-1	-2	0	3	-1	-4
E	-1	0	0	2	-4	2	5	-2	0	-3	-3	1	-2	-3	-1	0	-1	-3	-2	-2	1	4	-1	-4
G	0	-2	0	-1	-3	-2	-2	6	-2	-4	-4	-2	-3	-3	-2	0	-2	-2	-3	-3	-1	-2	-1	-4
H	-2	0	1	-1	-3	0	0	-2	8	-3	-3	-1	-2	-1	-2	-1	-2	-2	2	-3	0	0	-1	-4
I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4	2	-3	1	0	-3	-2	-1	-3	-1	3	-3	-3	-1	-4
L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4	-2	2	0	-3	-2	-1	-2	1	-4	-3	-1	-4	
K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5	-1	-3	-1	0	-1	-3	-2	-2	0	1	-1	-4
M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5	0	-2	-1	-1	-1	1	-3	-1	-1	-4	
F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6	-4	-2	-2	1	3	-1	-3	-3	-1	-4
P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7	-1	-1	-4	-3	-2	-2	-1	-2	-4
S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4	1	-3	-2	-2	0	0	0	-4
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	-2	-2	0	-1	-1	0	-4
W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11	2	-3	-4	-3	-2	-4
Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	-1	3	-2	-1	-4
V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4	-3	-2	-1	-4
B	-2	-1	3	4	3	0	1	-1	0	-3	-4	0	-3	-3	-2	0	-1	-4	-3	4	1	-1	-4	
Z	-1	0	0	1	-3	3	4	-2	0	-3	-3	1	-1	-3	-1	0	-1	-3	-2	-1	4	-1	-4	
X	0	-1	-1	-2	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-2	0	0	-2	-1	-1	-1	-1	-4	
*	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	1

BLOSUM62

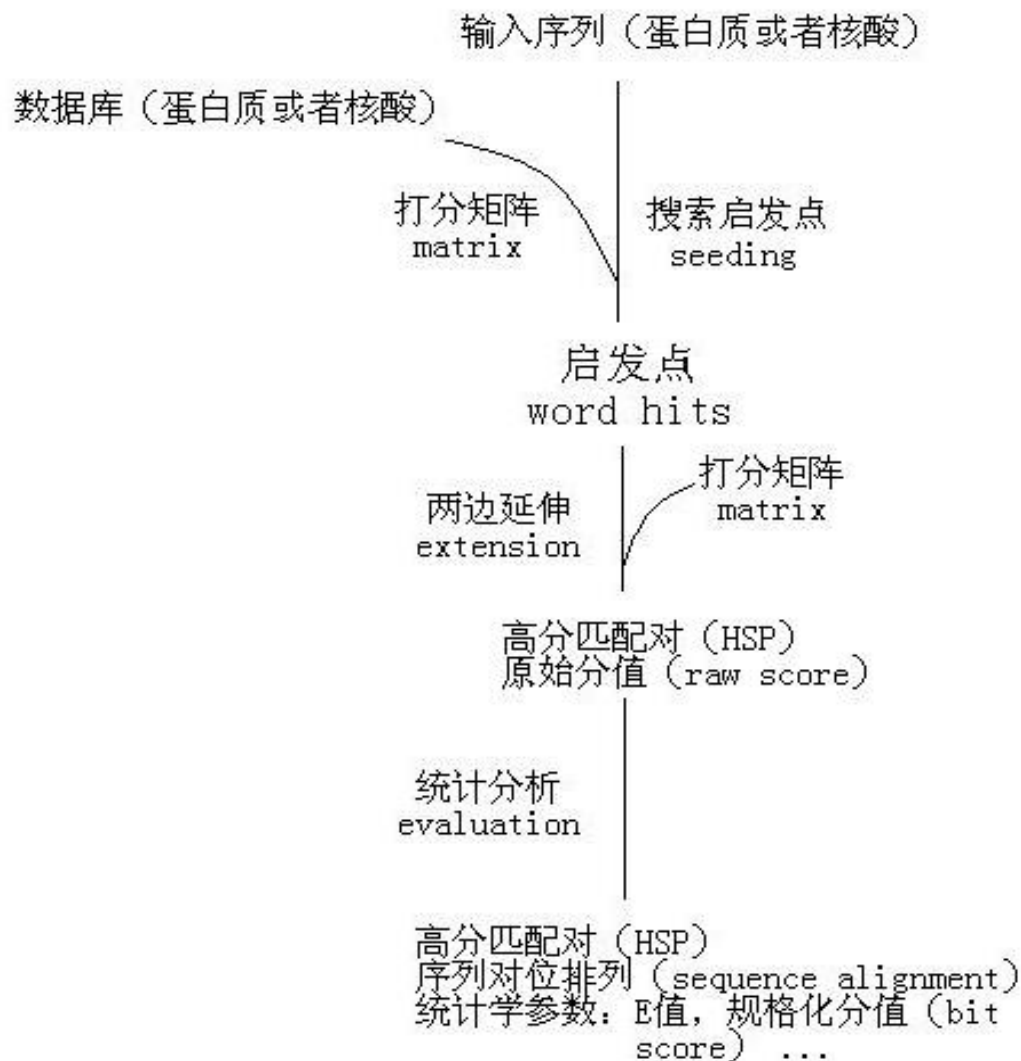
- BLAST算法得到的结果的统计显著性分析采用E-value。
- E-value: 期望值 (expect value)。它是指对一个给定的打分值在随机情况下在数据库中搜索比对的结果数目的期望。

$$E = kmne^{-\lambda S}$$

残基比例修正参数 打分系统修正参数 HSP分数

数据库中总残基数 查询序列残基数

Blast算法流程



Blast比对的5种组合

- All combinations (DNA/Protein) query and database.
 - DNA vs DNA → blastn
 - Protein vs Protein → blastp
 - DNA translation vs Protein → blastx
 - Protein vs DNA translation → tblastn
 - DNA translation vs DNA translation → tblastx

Frame（阅读框）

Frame1: 123123123123123123123

F2: - 12312312312312312312

F3: - - 1231231231231231231

正义链 ACTGCGTACGTACGATCAC

反向互补链 TGACGCATGCATGCATGTG

F-1: 321321321321321321321

F-2: 21321321321321321321-

F-3: 1321321321321321321- -

Nucleotide Translated BLAST Protein

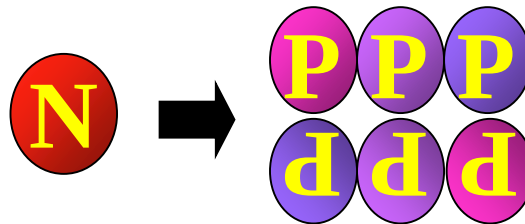
Particularly useful for nucleotide sequences without protein annotations, such as ESTs or genomic DNA

Program

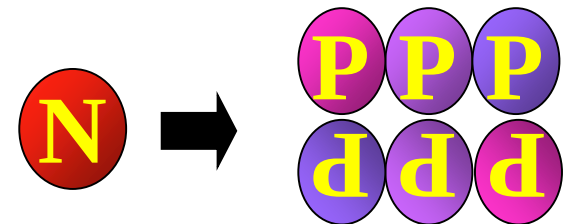
Query

Database

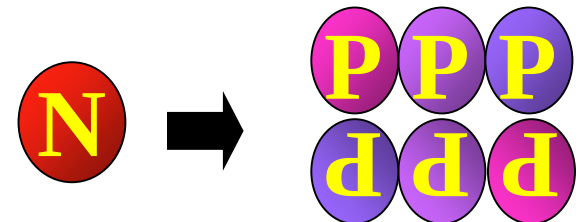
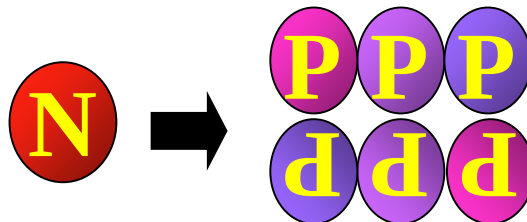
blastx



tblastn



tblastx

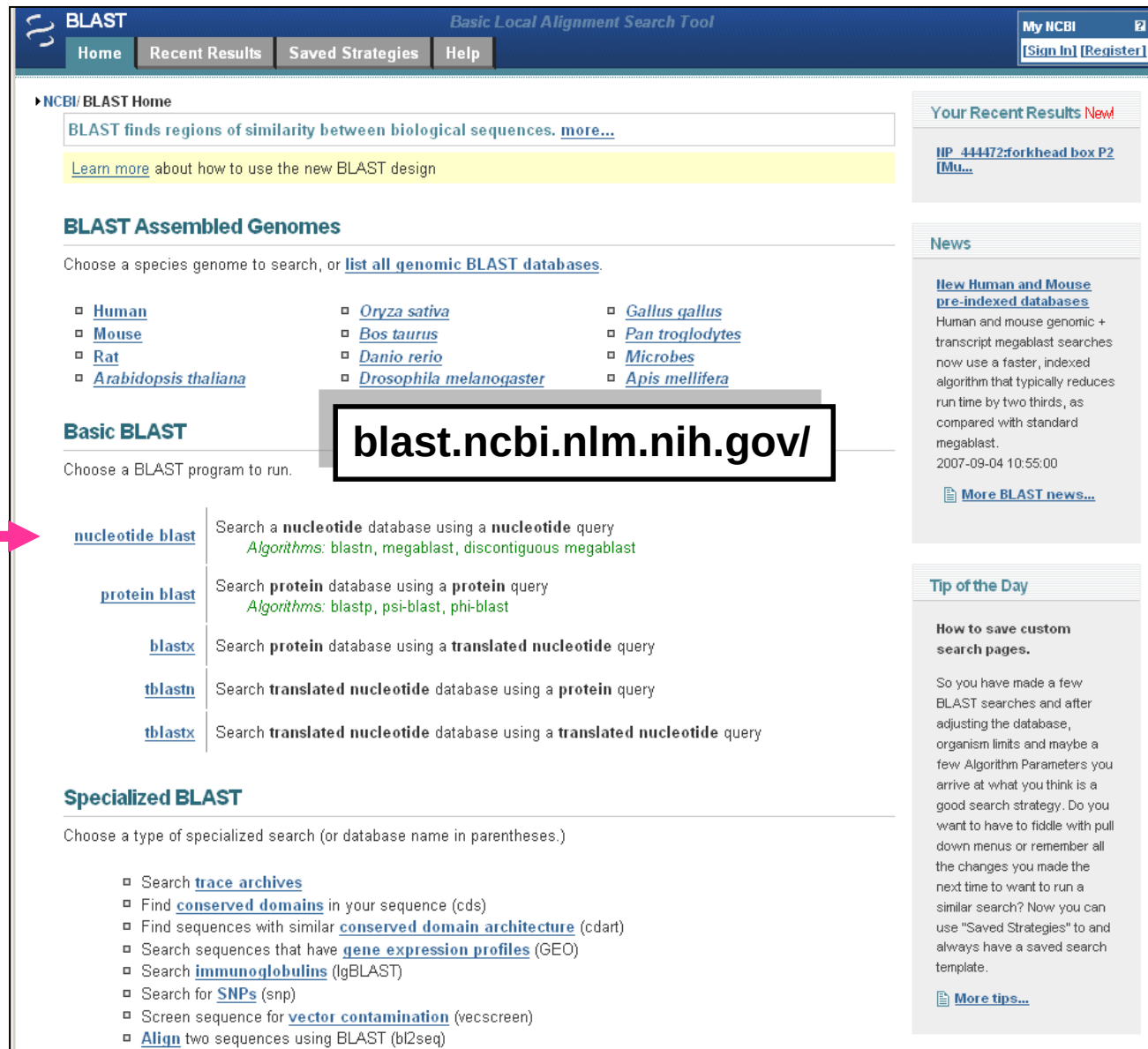


NCBI Molecular Biology Resources

Using NCBI BLAST

<http://blast.ncbi.nlm.nih.gov/>

The BLAST homepage



The screenshot shows the NCBI BLAST homepage. At the top, there's a navigation bar with 'Home', 'Recent Results', 'Saved Strategies', and 'Help'. A 'My NCBI' section includes 'Sign In' and 'Register' links. The main content area is titled 'NCBI/BLAST Home' and features a search bar with the text 'BLAST finds regions of similarity between biological sequences. more...'. Below this is a yellow banner with the text 'Learn more about how to use the new BLAST design'. The 'BLAST Assembled Genomes' section lists various species genomes for search, including Human, Mouse, Rat, Arabidopsis thaliana, Oryza sativa, Bos taurus, Danio rerio, Drosophila melanogaster, Gallus gallus, Pan troglodytes, Microbes, and Apis mellifera. The 'Basic BLAST' section provides a list of search programs: nucleotide blast, protein blast, blastx, tblastn, and tblastx, each with a brief description and the algorithms used. A pink arrow points to the 'nucleotide blast' link. The 'Specialized BLAST' section offers more advanced search options like trace archives, conserved domains, gene expression profiles, immunoglobulins, SNPs, vector contamination, and sequence alignment. On the right side, there are sections for 'Your Recent Results', 'News' (including a link to 'New Human and Mouse pre-indexed databases'), and 'Tip of the Day'.

blast.ncbi.nlm.nih.gov/

nucleotide blast Search a **nucleotide** database using a **nucleotide** query
Algorithms: blastn, megablast, discontinuous megablast

protein blast Search **protein** database using a **protein** query
Algorithms: blastp, psi-blast, phi-blast

blastx Search **protein** database using a **translated nucleotide** query

tblastn Search **translated nucleotide** database using a **protein** query

tblastx Search **translated nucleotide** database using a **translated nucleotide** query

Specialized BLAST

Choose a type of specialized search (or database name in parentheses.)

- Search [trace archives](#)
- Find [conserved domains](#) in your sequence (cds)
- Find sequences with similar [conserved domain architecture](#) (cdart)
- Search sequences that have [gene expression profiles](#) (GEO)
- Search [immunoglobulins](#) (IgBLAST)
- Search for [SNPs](#) (snp)
- Screen sequence for [vector contamination](#) (vecscreen)
- [Align](#) two sequences using BLAST (bl2seq)

Basic BLAST process

First: Enter the sequence

Home Recent Results Saved Strategies Help

NCBI/BLAST/blastn suite

blastn blastp blastx tblastn tblastx

Enter Query Sequence BLASTn programs search nucleotide databases using a nucleotide query. [more...](#)

Enter accession number, gi, or FASTA sequence [?](#) [Clear](#)

Query subrange [?](#)

From

To

Or, upload file [浏览...](#) [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database ☒ Human genomic + transcript ☐ Mouse genomic + transcript ☐ Others (nr etc.):

Human genomic plus transcript (Human G+T) [?](#)

Exclude [Optional](#)

☐ Models (XM/XP) ☐ Environmental sample sequences

Entrez Query [Optional](#)

Enter an Entrez query to limit search [?](#)

Basic BLAST process

Second: Select Databases and program

Choose Search Set

Database
Human genomic + transcript (selected) Mouse genomic + transcript Others (nr etc.):
Human genomic plus transcript (Human G+T) [v] [?]

Exclude
Optional
☐ Models (XM/XP) ☐ Environmental sample sequences

Entrez Query
Optional
[]
Enter an Entrez query to limit search [?]

Program Selection

Optimize for
Highly similar sequences (megablast) (selected)
More dissimilar sequences (discontiguous megablast)
Somewhat similar sequences (blastn)
Choose a BLAST algorithm [?]

BLAST [?] Search database Human G+T using Megablast (Optimize for highly similar sequences)
☐ Show results in a new window

► [Algorithm parameters](#)

Third: BLAST button

Nucleotide Databases: Traditional

The screenshot displays the NCBI BLAST search interface. The 'Choose Search Set' dropdown menu is open, showing a list of databases. The 'Human genomic plus transcript (Human G+T)' option is selected. The menu is divided into two sections: 'Genomic plus Transcript' and 'Other Databases'. The 'Other Databases' section lists various sequence collections, including Nucleotide collection (nr/nt), Reference mRNA sequences (refseq_rna), Reference genomic sequences (refseq_genomic), NCBI Genomes (chromosome), Expressed sequence tags (est), Non-human, non-mouse ESTs (est_others), Genomic survey sequences (gss), High throughput genomic sequences (HTGS), Patent sequences (pat), Protein Data Bank (pdb), Human ALU repeat elements (alu_repeats), Sequence tagged sites (dbsts), Whole-genome shotgun reads (wgs), and Environmental samples (env_nt). The 'BLAST' button is visible at the bottom left, and the 'Algorithm parameters' link is at the bottom right.

Choose Search Set

Database

☒ Human genomic + transcript ☐ Mouse genomic + transcript ☐ Others (nr etc.):

Human genomic plus transcript (Human G+T)

Genomic plus Transcript

- Human genomic plus transcript (Human G+T)
- Mouse genomic plus transcript (Mouse G+T)

Other Databases

- Nucleotide collection (nr/nt)
- Reference mRNA sequences (refseq_rna)
- Reference genomic sequences (refseq_genomic)
- NCBI Genomes (chromosome)
- Expressed sequence tags (est)
- Non-human, non-mouse ESTs (est_others)
- Genomic survey sequences (gss)
- High throughput genomic sequences (HTGS)
- Patent sequences (pat)
- Protein Data Bank (pdb)
- Human ALU repeat elements (alu_repeats)
- Sequence tagged sites (dbsts)
- Whole-genome shotgun reads (wgs)
- Environmental samples (env_nt)

Exclude

Optional

Entrez Query

Optional

Program Selection

Optimize for

BLAST

☐ Show results in a new window

[Algorithm parameters](#)

Services
blastn
tblastn
tblastx

Nucleotide Databases:

Traditional

Databases are mostly non-overlapping

- **nr (nt)**
 - Traditional GenBank
 - NM_ and XM_ RefSeqs
 - [refseq_rna](#)
- **refseq_genomic**
 - NC_ RefSeqs
- **dbest**
 - EST Division
 - [est_human](#), [mouse](#), [others](#)

- **htgs**
 - HTG division
- **gss**
 - GSS division
- **wgs**
 - whole genome shotgun
- **env nt**
 - environmental samples

Basic BLAST process

The screenshot displays the NCBI BLAST search interface. The 'Choose Search Set' section is active, showing a dropdown menu with the following options:

- Genomic plus Transcript**
 - Human genomic plus transcript (Human G+T) (selected)
 - Mouse genomic plus transcript (Mouse G+T)
- Other Databases**
 - Nucleotide collection (nr/nt)
 - Reference mRNA sequences (refseq_rna)
 - Reference genomic sequences (refseq_genomic)
 - NCBI Genomes (chromosome)
 - Expressed sequence tags (est)
 - Non-human, non-mouse ESTs (est_others)
 - Genomic survey sequences (gss)
 - High throughput genomic sequences (HTGS)
 - Patent sequences(pat)
 - Protein Data Bank (pdb)
 - Human ALU repeat elements (alu_repeats)
 - Sequence tagged sites (dbsts)
 - Whole-genome shotgun reads (wgs)
 - Environmental samples (env_nt)

Other visible elements include:

- Database:** Radio buttons for 'Human genomic + transcript' (selected), 'Mouse genomic + transcript', and 'Others (nr etc.):'.
- Exclude:** 'Optional' and 'Entrez Query' (Optional).
- Program Selection:** 'Optimize for'.
- BLAST** button.
- Show results in a new window** checkbox.
- Algorithm parameters** link.



BLAST

Search **database Human G+T** using **Megablast (Optimize for highly similar sequences)**

☐ Show results in a new window

▼ **Algorithm parameters**

General Parameters

Max target
sequences

100 ▼

Select the maximum number of aligned sequences to display ?

Short queries

☒ Automatically adjust parameters for short input sequences ?

Expect threshold

10 ?

Word size

28 ▼ ?

Word Size

Scoring Parameters

Match/Mismatch
Scores

1,-2 ▼ ?

Gap Costs

Linear ▼ ?

Filters and Masking

Filter

☐ Low complexity regions ?
☐ Species-specific repeats for: Human ▼ ?

Mask

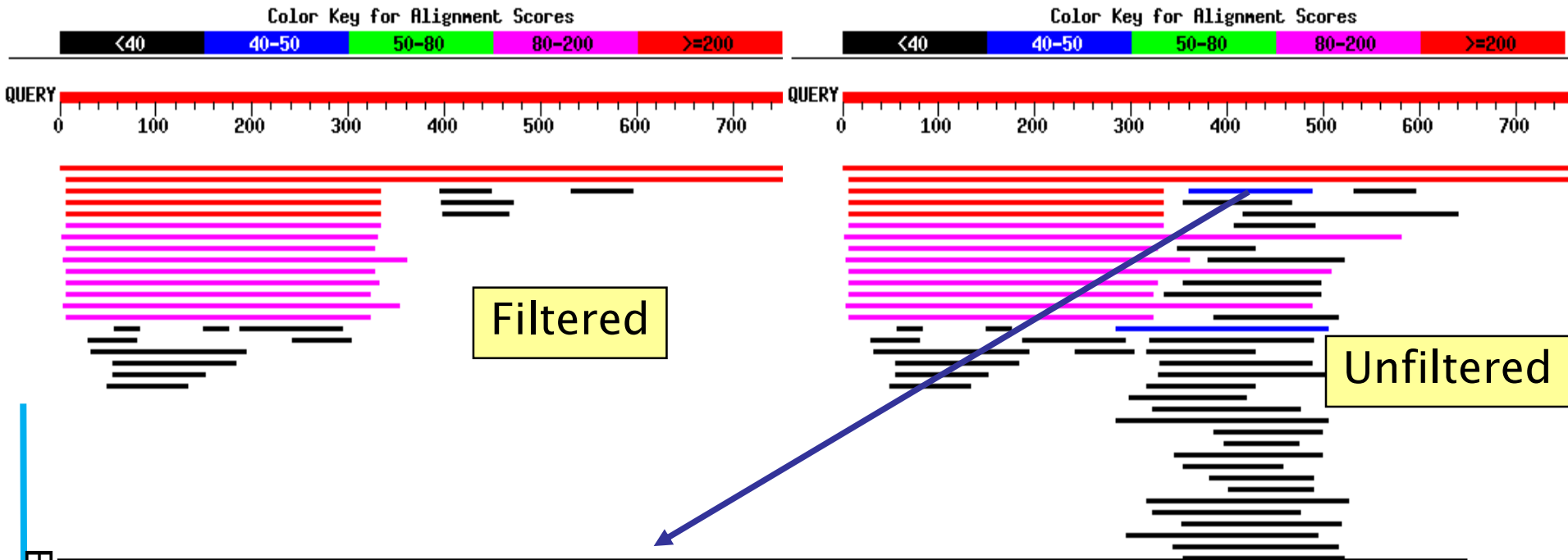
☒ Mask for lookup table only ?
☐ Mask lower case letters ?

BLAST

Search **database Human G+T** using **Megablast (Optimize for highly similar sequences)**

☐ Show results in a new window

Low Complexity Filtering



sp|P27476|NSR1_YEAST NUCLEAR LOCALIZATION SEQUENCE BINDING PROTEIN (P67)

Length = 414

Score = 40.2 bits (92), Expect = 0.013

Identities = 35/131 (26%), Positives = 56/131 (42%), Gaps = 4/131 (3%)

Query: 362 STTSLTSSSTSGSSDKVYAHQMVRTDSREQKLDAFLQPLSKPLS---SQPQAIVTEDKTD 418

S++S SSS+S SS + + ++S + + S S S+ + E K

Sbjct: 29 SSSSESSSSSSSSSSSESESESESESESSSSSSSSSDSESSSSSSSDSESEAEATKKEESKDS 88

Basic BLAST: Protein Searches

Protein BLAST Page

Enter Query Sequence

Enter accession number, gi, or FASTA sequence ? [Clear](#)

>Mutated in Colon Cancer
IETVYAAYLPKNTHPFLYLSLEISPQNVDVNVHPTKHEVHFLHEESILERVQQHIESKLLGS
NSSRMFTQTLLPGLAGPSGEMVKSTTSLTSSSTSGSSDKVYAHQMVRTDSREQKLDNFLQP
LSKPLSS

Query subrange ?
From
To

Or, upload file [Browse...](#) ?

Job Title

Enter a descriptive title for your BLAST search ?

Choose Search Set

Database

Organism
Optional

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. ?

Entrez Query
Optional

Enter an Entrez query to limit search ?

Bioinformatics

BLAST Databases: Non-redundant protein

Choose Search Set

Database

Non-redundant protein sequences (nr) ?

Non-redundant protein sequences (nr)

Reference proteins (refseq_protein)

Organism

Optional

Entrez Q

Optional

Services
blastp
blastx

nr (non-redundant protein sequences)

- GenBank CDS translations
- NP_, XP_ **RefSeqs**
- Outside Protein
 - PIR, **Swiss-Prot**, PRF
 - **PDB** (sequences from structures)

pat protein patents

env nr environmental samples

Limiting Database: Organism

Choose Search Set

Database

Organism
Optional

Entrez Query
Optional

bacter

CFB group **bacteria** (taxid:976)
GNS **bacteria** (taxid:200795)
green sulfur **bacteria** (taxid:1090)
Bacteria (taxid:2)
purple **bacteria** and relatives (taxid:1224)
purple non-sulfur **bacteria** (taxid:1224)
purple photosynthetic **bacteria** (taxid:1224)
purple photosynthetic **bacteria** and relatives (taxid:1224)
purple **bacteria** (taxid:1224)
low G+C Gram-positive **bacteria** (taxid:1239)

Organism autocomplete

Limiting Database: Entrez Query

Entrez Query
Optional

all[filter] NOT mammals[organism]

Enter an Entrez query to limit search 

all[filter] NOT mammals[organism]

gene_in_mitochondrion[Properties]
2006:2007 [Modification Date]

Nucleotide

biomol_mrna[Properties]

biomol_genomic[Properties]

Run Search

The screenshot shows the NCBI BLAST web interface. The 'Program Selection' tab is active, displaying three radio button options: 'blastp (protein-protein BLAST)' (selected), 'PSI-BLAST (Position-Specific Iterated BLAST)', and 'PHI-BLAST (Pattern Hit Initiated BLAST)'. Below these is a link 'Choose a BLAST algorithm'. The 'BLAST' button is highlighted with a large blue arrow. The search database is set to 'swissprot' and the algorithm is 'Blastp (protein-protein BLAST)'. The checkbox 'Show results in a new window' is unchecked. A yellow note at the bottom right states: 'Note: Parameter values that differ from the default are highlighted in yellow'. The 'Algorithm parameters' link is visible at the bottom left.

Program Selection

Algorithm

- ☒ blastp (protein-protein BLAST)
- ☐ PSI-BLAST (Position-Specific Iterated BLAST)
- ☐ PHI-BLAST (Pattern Hit Initiated BLAST)

Choose a BLAST algorithm [?](#)

BLAST

Search database **swissprot** using **Blastp (protein-protein BLAST)**

☐ Show results in a new window

[Algorithm parameters](#)

Note: Parameter values that differ from the default are highlighted in yellow

BLAST Formatting Page

► [NCBI/BLAST](#) / Formatting Results - 1S1DCBME015 [\[Formatting options\]](#)

Job Title: Mutated in colon cancer

Putative conserved domains have been detected, click on the image below for detailed results.



WAITING

Request ID	1S1DCBME015
Status	Searching
Submitted at	Tue Apr 10 17:46:20 2007
Current time	Tue Apr 10 17:46:57 2007
Time since submission	00:00:36

Conserved Domain Results

This page will be automatically updated in **10** seconds until search is done

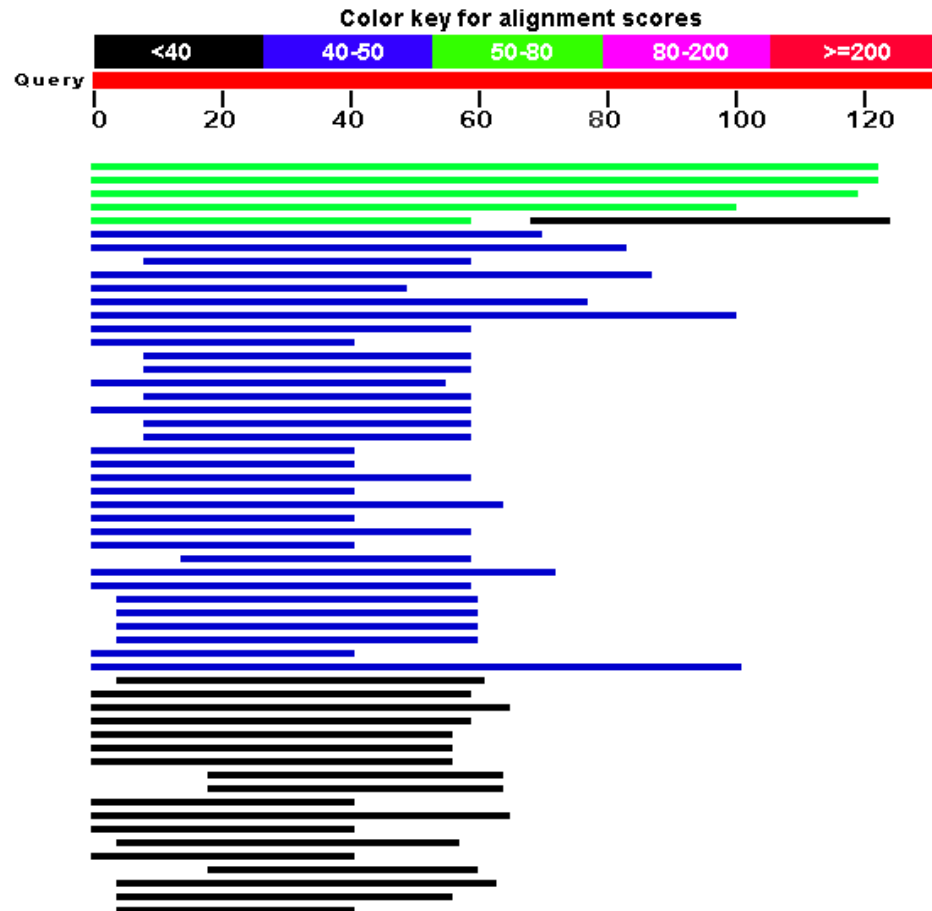
BLAST Output: Graphical Overview

[Taxonomy reports](#)

Query= Mutated in colon cancer
Length=131

[Distribution of 78 Blast Hits on the Query Sequence](#)

P38920 DNA mismatch repair protein MLH1 (MutL protein homolog .. S=74 E=4.9e-14



BLAST Output: Descriptions

Sequences producing significant alignments			Sorted by e values	Score (Bits)	E Value	
sp P38920 MLH1	YEAST	DNA mismatch repair protein MLH1 (MutL p...		74.7	5e-14	G
sp Q9P7W6 MLH1	SCHPO	Putative MutL protein homolog 1 (DNA mis...		74.7	6e-14	
sp Q8RA70 MUTL	THETN	DNA mismatch repair protein mutL		59.7	2e-09	
sp Q72PF7 MUTL	LEPIC	DNA mismatch repair protein mutL		53.9	1e-07	
sp P44494 MUTL	AEIN	DNA mismatch repair protein mutL		53.1	2e-07	
sp P57886 MUTL	SMU	DNA mismatch repair protein mutL		48.1	4e-06	
sp Q88DD1 MUTL	K	DNA mismatch repair protein mutL		47.1	6e-06	G
sp P74925 MUTL		DNA mismatch repair protein mutL		47.8	7e-06	
sp Q9KAC1 MUTL		DNA mismatch repair protein mutL		46.6	2e-05	
sp Q97I20 MUTL		DNA mismatch repair protein mutL		46.2	2e-05	
sp Q821I9 MUTL	CHLCV	DNA mismatch repair protein mutL		46.2	2e-05	
sp Q8KAX3 MUTL	CHLTE	DNA mismatch repair protein mutL		45.8	2e-05	
sp Q87AC9 MUTL	XYLFT	DNA mismatch repair protein mutL		45.8	3e-05	
sp Q1RGP6 MUTL	RICBR	DNA mismatch repair protein mutL		45.1	5e-05	
sp Q8XDN4 MUTL	ECO57	DNA mismatch repair protein mutL		44.7	6e-05	
sp Q8FAK9 MUTL	ECOL6	DNA mismatch repair protein mutL		44.7	7e-05	
sp Q5E2C6 MUTL	VIBF1	DNA mismatch repair protein mutL		44.7	7e-05	G

Link to entrez

5 X 10⁻¹⁴

Gene Linkout

Default e value cutoff 10

sp P0A3R1 HEXB	STRPN	DNA mismatch repair protein mutL	sp P0A...	35.8	0.032
sp P57633 MUTL	BUCAI	DNA mismatch repair protein mutL		35.4	0.036
sp Q9Z794 MUTL	CHLPN	DNA mismatch repair protein mutL		34.7	0.058
sp O67518 MUTL	AQUAE	DNA mismatch repair protein mutL		34.3	0.074
sp Q8K913 MUTL	BUCAP	DNA mismatch repair protein mutL		33.9	0.095
sp Q9HSM6 MUTL	HALSA	DNA mismatch repair protein mutL		33.1	0.19
sp Q06R72 YCF1	JASNU	Putative membrane protein ycf1		32.3	0.30
sp P73349 MUTL	SYNY3	DNA mismatch repair protein mutL		31.2	0.78
sp Q5ZXU2 DEOB	LEGPH	Phosphopentomutase (Phosphodeoxyribomutase)		29.6	1.9
sp P02239 LGB1	LUPLU	Leghemoglobin-1 (Leghemoglobin I)		29.3	2.5
sp P02240 LGB2	LUPLU	Leghemoglobin-2 (Leghemoglobin II)		27.7	7.7

BLAST Output: Alignments

```
>sp|P23367|MUTL ECOLI DNA mismatch repair protein mutL  
Length=615
```

```
Score = 44.7 bits (104), Expect = 7e-05, Method: Composition-based stats.  
Identities = 25/59 (42%), Positives = 33/59 (55%), Gaps = 8/59 (13%)
```

```
Query 9 LPKNTHPFLYLSLEIS PQNV DVNVHP TKHEVHF-----LHE---ESILERVQQHIESKL 59  
      L + P L LEI P VDVNVHP KHEV F +H+ + +L +QQ +E+ L  
Sbjct 280 LGADQQPAFVLYLEIDPHQVDVNVHPAKHEVRFHQ SRLVHDFIYQGVLSVLQQQLETP L 338
```

Identical match

positive score
(conservative)

Negative or zero

gap

[NCBI/BLAST/blastp suite/Formatting Results - GTV2VJET015](#)[Edit and Resubmit](#)[Save Search Strategies](#)[▶Formatting options](#)[▶Download](#)**pdb|1cbh| (36 letters)**

Query ID [gi|229733|pdb|1CBH|](#)
Description None
Molecule type amino acid
Query Length 36

Database Name pdb
Description PDB pro
Program BLASTP

Other reports: [▶Search Summary](#) [\[Taxonomy reports\]](#) [\[Distance tree of results\]](#) [\[Related Structures\]](#) [\[Multiple alignment\]](#) **NEW**

▼ Graphic Summary[▼ Show Conserved Domains](#)

Putative conserved domains have been detected, click on the image below for details

[\[Taxonomy reports\]](#)**Distribution of 5 Blast Hits on the Query Sequence ⓘ**

Mouse over to see the define, click to show alignments

Color key for alignment scores[\[Multiple alignment\]](#) **NEW**

TaxBLAST: Taxonomy Reports

Organism Report

[Saccharomyces cerevisiae](#) (yeast, ...) [[ascomycetes](#)] taxid 4932
[sp|P38920|MLH1 YEAST](#) DNA mismatch repair protein MLH1 (Mut... 74 5e-14

[Schizosaccharomyces pombe](#) [[ascomycetes](#)] taxid 4896
[sp|Q9P7W6|MLH1 SCHPO](#) Putative MutL protein homolog 1 (DNA ... 74 6e-14
[sp|P54280|PMS1 SCHPO](#) DNA mismatch repair protein pms1 36 0.014

[Thermoanaerobacter tengcongensis](#) [[eubacteria](#)] taxid 119072
[sp|Q8RA70|MUTL THETN](#) DNA mismatch repair protein mutL 59 2e-09

[Leptospira interrogans](#) [[eubacteria](#)] taxid 44275
[sp|Q72PF7|MUTL LEPIC](#) DNA mismatch repair protein mutL 44 7e-05

[Escherichia coli](#) [[enterobacteria](#)] taxid 562
[sp|P23367|MUTL ECOLI](#) DNA mismatch repair protein mutL 44 7e-05

[Haemophilus ducreyi](#) [[g-proteobacteria](#)] taxid 730
[sp|Q7VN43|MUTL HAEDU](#) DNA mismatch repair protein mutL 44 8e-05

[Salmonella typhi](#) [[enterobacteria](#)] taxid 601
[sp|Q8Z187|MUTL SALT1](#) DNA mismatch repair protein mutL 43 1e-04

[Salmonella typhimurium](#) [[enterobacteria](#)] taxid 602
[sp|P14161|MUTL SALTY](#) DNA mismatch repair protein mutL 43 1e-04

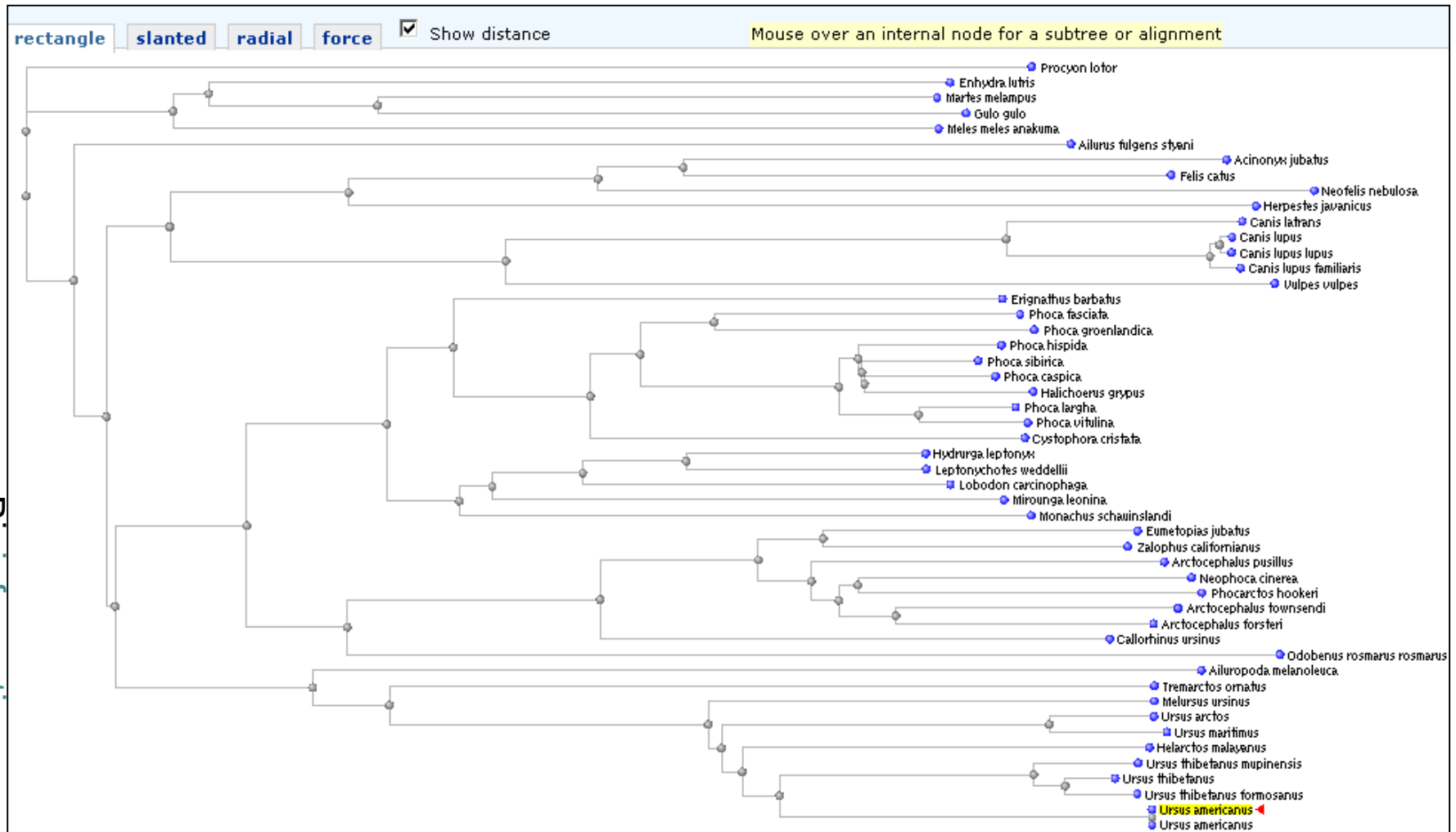
[Pseudomonas syringae pv. tomato](#) [[g-proteobacteria](#)] taxid 323
[sp|Q87VJ2|MUTL PSESM](#) DNA mismatch repair protein mutL 43 1e-04

[Pseudomonas aeruginosa](#) [[g-proteobacteria](#)] taxid 287
[sp|Q9HUL8|MUTL PSEAE](#) DNA mismatch repair protein mutL 43 1e-04

[Xylella fastidiosa](#) [[g-proteobacteria](#)] taxid 2371
[sp|Q9PFB8|MUTL XYLFA](#) DNA mismatch repair protein mutL 42 2e-04

[Rickettsia felis](#) [[a-proteobacteria](#)] taxid 42862
[sp|Q4UJQ1|MUTL RICFE](#) DNA mismatch repair protein mutL 42 2e-04

TaxBLAST: Distance tree of results



Nucleotide Translated BLAST Protein

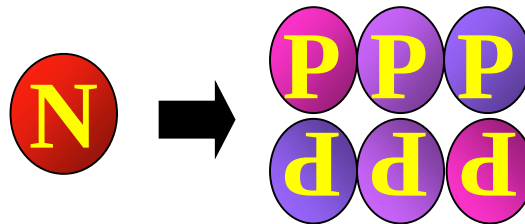
Particularly useful for nucleotide sequences without protein annotations, such as ESTs or genomic DNA

Program

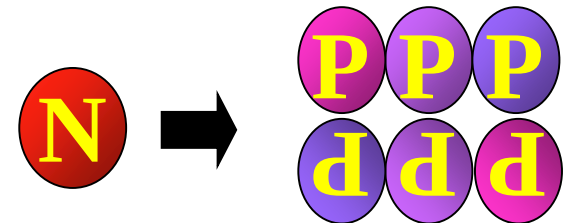
Query

Database

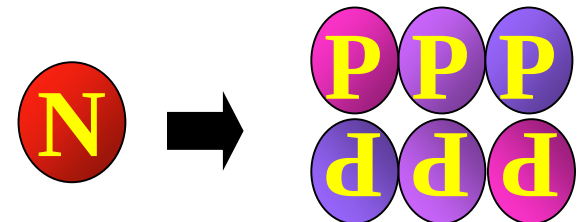
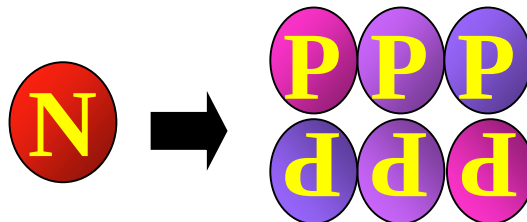
blastx



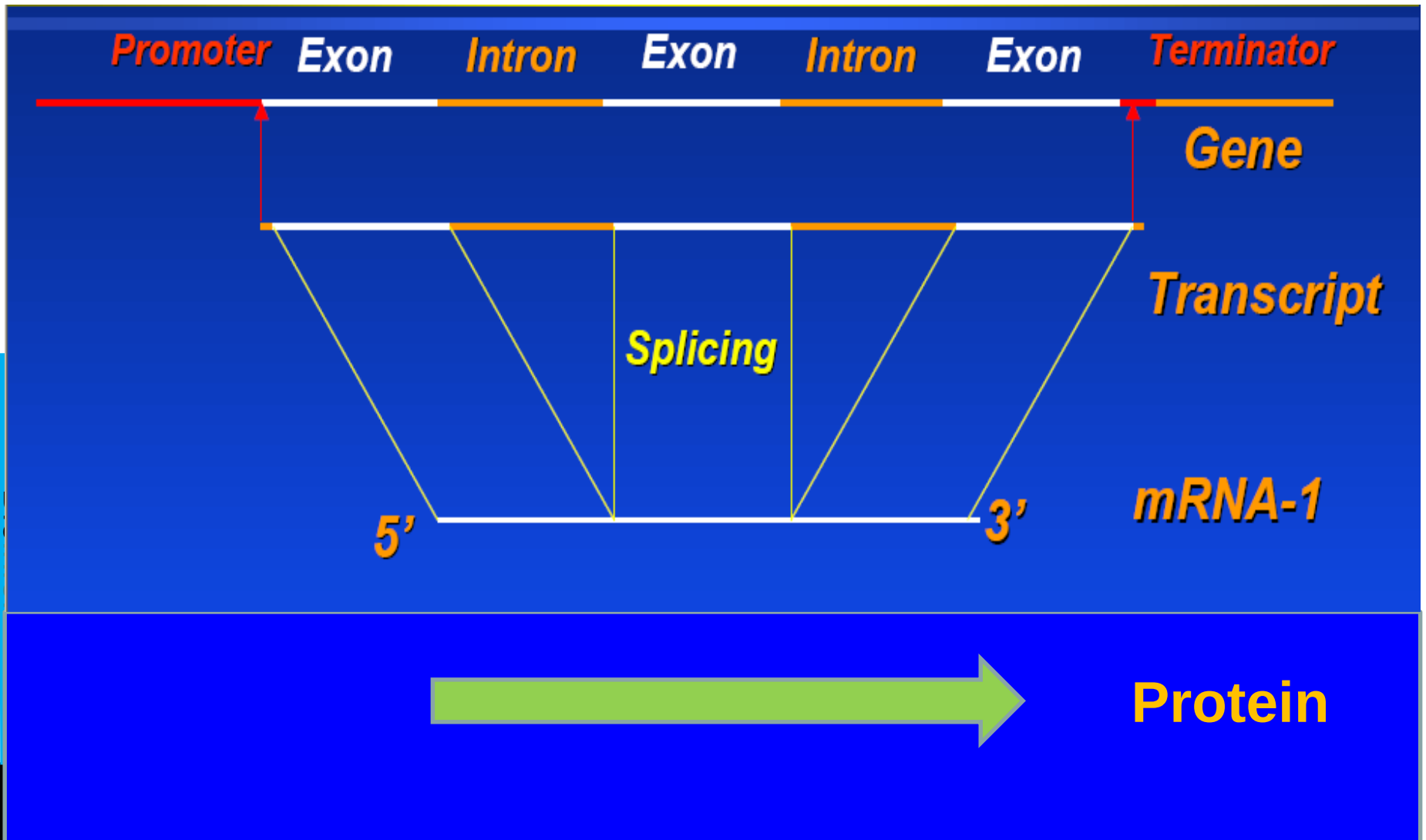
tblastn



tblastx



tBlastn/BlastX



Poplar Genome BLAST

NCBI Home ► Genomic Biology ► BLAST

Search Map Viewer

BLAST
Overview
FAQs
News
Manual
References
Retrieve results
Genome Project

BLAST Black cottonwood Sequences

☒ Enter an accession, gi, or a sequence in FASTA format:

MEACNCIEPQWPADELLMKYQYISDFFIALAYFSIPLELIYFVKKS AVFPYRWVLVQFGAFIVLC
INLWTFMSHSRTVAIVMTTAKVLTAVVSCATALMLVHIIPDLLSVKTRELF LKNKAAELDREMG
ETGRHVRMLTHEIRSTLDRHTILKTTLVELGRTLALEECALWMPTRTGLELQLSYTLROONPVGY
LPVINQVPSSNRALKISPNSPVARMRLAGKHMPGEVVAVRVPLHLHLSNFPQINDWPELSTKRYAL
PSDSARQWHVHELELVEVVADQVAVALSHAAILEESMRARDLLMEQNIALDLARREATAIRARN

☐ Or, choose a file to upload

Set subsequence: (optional)
From: To:

Database:
 45688 sequences

Program:

Optional
Expect

Advanced options:

genomic (reference only)
HTGS
Non-RefSeq RNA
Non-RefSeq protein
ESTs
Traces-other*
WGS contigs

megaBLAST: Compare highly related nucleotide sequences
cross-species megaBLAST: Compare nucleotide sequences for other species to this genome
BLASTN: Compare nucleotide sequences
BLASTP: Compare protein sequences
BLASTX: Compare a nucleotide sequence against a protein database
TBLASTN: Compare a protein sequence against a nucleotide database

tblastn Genome BLAST Results

Database: Populus trichocarpa genomic (reference only)
45,688 sequences; 880,289,017 total letters

If you have any problems or questions with the results of this search
please refer to the [BLAST FAQs](#)
[Taxonomy reports](#)

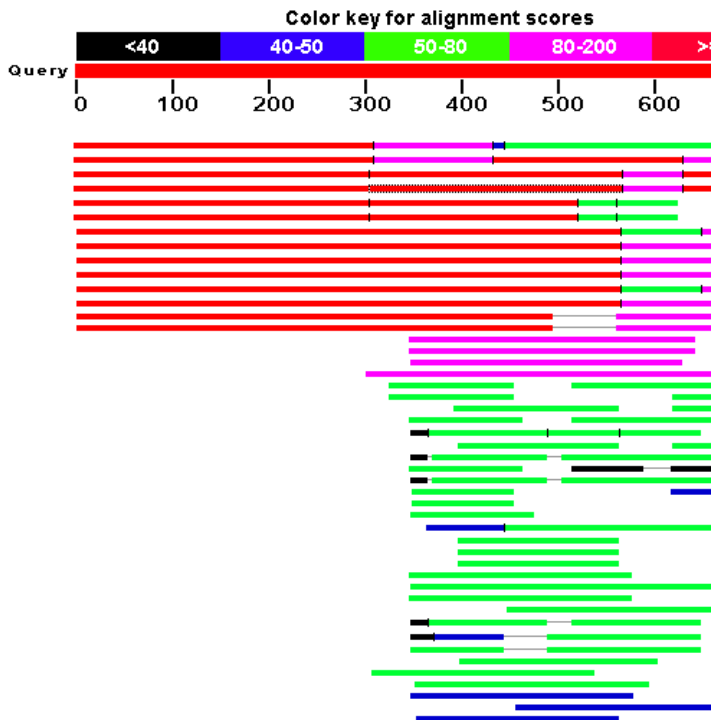
Genome View

Show positions of the BLAST hits in the black cottonwood genome using the

Query= gi|6841075|gb|AAF28893.1|AF124527_1 ethylene receptor [Prunus persica]
Length=738

Distribution of 103 Blast Hits on the Query Sequence

NC_008467 Populus trichocarpa linkage group I, whole genome shotg.. S=194 E=1.1



Protein-nucleotide alignments

Score = 284 bits (728), Expect = 1e-164, Identities = 164/283 (57%), Positives = 177/283 (62%), Gaps = 84/283 (29%)
Frame = -1

Query	Subject	Score	Expect	Identities	Positives	Gaps
425	EVHNLIKPVASVKKLSVSLNLAADLPVQAVGDEKRLMQIVLNVVGNVAVKFSKEGSISITA	484				
12443459	+V NLIKPVASVKKLPVTLNLAADLPVQAVGDEKRLMQIVLNVVGNVAVKFSKEGSISITA	12443280				
485	FVAKSESRLDFRAPEFFPAQSDNHFYLRVQ-----	514				
12443279	FVAKSESRLDFRAPEFFPAQSDNHFYLRVQ-----	12443100				
515	--VKDSGSGINPQDIPKLFKFAQTQSLATRNSSGSGGLAICK-----	556				
12443099	--VKDSGSGINPQDIPKLFKFAQTQSLATRNSSGSGGLAICK-----	12442920				
557	-----RFVNLMEGHIWIESEGPCKGCTAIFI	582				
12442919	-----RFVNLMEGHIWIESEGPCKGCTAIFI	12442740				
583	VKLGFAERSNESKLPFLTKVQANHVQTNFPGKVLVMDNGSV	625				
12442739	VKLGFAERSNESKLPFLTKVQANHVQTNFPGKVLVMDNGSV	12442617				

Exons and genes mixed

Score = 194 bits (493), Expect = 1e-114, Identities = 114/130 (87%), Positives = 124/130 (95%), Gaps = 2/130 (1%)
Frame = -3

Query	Subject	Score	Expect	Identities	Positives	Gaps
300	ADQVAVALSHAAILEESMRARDLLMEQNIALDLarreaetairarNDFLAVMNHMERTPM	359				
12443955	ADQVAVALSHAAILEESMRARDLLMEQNIALDLarreaetairarNDFLAVMNHMERTPM	12443776				
360	HAIALLSSLLQETELTPEQRLMVETILKSSHLLATLINDVLDLSRLEDGSLQLEIATFNL	419				
12443775	HAIALLSSLLQETELTPEQRLMVETILKSSHLLATLINDVLDLSRLEDGSLQLEIATFNL	12443596				
420	HSVFREVHNL 429					
12443595	HSVFREVHNL 429	12443566				

Score = 186 bits (473), Expect = 1e-45, Identities = 95/114 (83%), Positives = 102/114 (89%), Gaps = 2/114 (1%)
Frame = -3

Query	Subject	Score	Expect	Identities	Positives	Gaps
625	VTKGLLVHLGCDVTTVSSIDEFLHVISQEHKVVFMVCMPIGDGYELAVRIHEKFTKRHE	684				
12440235	VTKGLLVHLGCDVTTVSSIDEFLHVISQEHKVVFMVCMPIGDGYELAVRIHEKFTKRHE	12440062				
685	RPVLVALTGNIDKMKENCMRVGMDGVILKPVSVDKMRSVLSLEHRLVFEAM	738				
12440061	RPVLVALTGNIDKMKENCMRVGMDGVILKPVSVDKMRSVLSLEHRLVFEAM	12439900				

Specialized BLAST Pages

Specialized BLAST

Choose a type of specialized search (or database name in parentheses.)

- ▣ Make specific primers with [Primer-BLAST](#)
- ▣ Search [trace archives](#)
- ▣ Find [conserved domains](#) in your sequence (cds)
- ▣ Find sequences with similar [conserved domain architecture](#) (cdart)
- ▣ Search sequences that have [gene expression profiles](#) (GEO)
- ▣ Search [immunoglobulins](#) (IgBLAST)
- ▣ Search for [SNPs](#) (snp)
- ▣ Screen sequence for [vector contamination](#) (vecscreen)
- ▣ [Align](#) two sequences using BLAST (bl2seq)
- ▣ Search [protein](#) or [nucleotide](#) targets in PubChem BioAssay

本地版Blast

- 可通过NCBI的ftp站点获得，有适合不同平台的版本（包括linux，dos等）。
- 获得程序的同时必须获取相应的数据库才能在本地进行blast分析。
- 单机版的优点是可以处理大批的数据，可以自己定义数据库
- 需要一定的计算机操作水平。

单机版的Blast使用（一）

为什么使用单机版的Blast？

- 1.特殊的数据库要求。
- 2.涉及序列的隐私与价值。
- 3.批量处理

单机版的Blast使用（二）

单机版Blast的基本操作过程

1. 下载单机版的Blast程序

<ftp://ftp.ncbi.nlm.nih.gov/blast/executables/>

目录下，下载对应的操作系统版本。

2. 解压程序包(blast-2.28-ia32-linux.tar.gz)

命令是：

```
$ tar zxvf blast-2.28-ia32-linux.tar.gz
```

下载正确的Blast程序包

blast:在本地运行的blast程序包

wwwblast:在本地服务器建立blast服务的网站

netblast:blast的客户端程序，直接链接至NCBI的BLAST服务器，使用BLAST服务，不需浏览器。

返回

BLAST Basic Local Alignment Search Tool

Home Recent Results Saved Strategies Help

NCBI/ BLAST/ blastn suite

blastn blastp blastx tblastn tblastx

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) ?

Or, upload file

Job Title

Enter a descriptive title for your BLAST search ?

☐ Align two or more sequences ?

Choose Search Set

Database

☒ Human genomic + transcript ☐ Mouse genomic + transcript

Human genomic plus transcript (Human G+T) ?

Exclude Optional

☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Entrez Query Optional

Enter an Entrez query to limit search ?

Program Selection

ENTER
Sequences
Here

Select
Database

单机版的Blast使用（三）

3. 获取Blast数据库

a. 直接从ncbi下载

<ftp://ftp.ncbi.nlm.nih.gov/blast/db/>

b. 用Blast程序包提供的
formatdb工具自己格 式化序列
数据成数据库。

假设有一序列数据

（sequence.fa，多序列，fasta
格式），欲自己做成Blast数据
库，典型的命令如下：

← → ↻ ① 不安全 | ftp://ftp.ncbi.nlm.nih.gov/blast/db/

/blast/db/ 的索引

 [上级目录]

名称	大小	修改日期
 16SMicrobial.tar.gz	32.3 MB	2019/4/14 下午12:04:00
 16SMicrobial.tar.gz.md5	54 B	2019/4/14 下午12:04:00
 FASTA/		2019/4/19 上午11:50:00
 README	0 B	2011/7/12 上午8:00:00
 cdd_delta.tar.gz	1.5 GB	2019/4/2 下午1:03:00
 cdd_delta.tar.gz.md5	51 B	2019/4/2 下午1:04:00
 cloud/		2019/4/19 下午12:00:00
 env_nr.00.tar.gz	822 MB	2019/4/2 上午8:06:00
 env_nr.00.tar.gz.md5	51 B	2019/4/2 上午8:07:00
 env_nr.01.tar.gz	381 MB	2019/4/2 上午8:07:00
 env_nr.01.tar.gz.md5	51 B	2019/4/2 上午8:08:00
 env_nt.00.tar.gz	869 MB	2019/4/2 下午7:18:00
 env_nt.00.tar.gz.md5	51 B	2019/4/2 下午7:18:00
 env_nt.01.tar.gz	1.0 GB	2019/4/2 下午7:20:00
 env_nt.01.tar.gz.md5	51 B	2019/4/2 下午7:20:00
 env_nt.02.tar.gz	814 MB	2019/4/2 下午7:22:00
 env_nt.02.tar.gz.md5	51 B	2019/4/2 下午7:22:00
 env_nt.03.tar.gz	1.1 GB	2019/4/2 下午7:24:00
 env_nt.03.tar.gz.md5	51 B	2019/4/2 下午7:24:00
 env_nt.04.tar.gz	1.1 GB	2019/4/2 下午7:26:00
 env_nt.04.tar.gz.md5	51 B	2019/4/2 下午7:26:00
 env_nt.05.tar.gz	1.1 GB	2019/4/2 下午7:27:00
 env_nt.05.tar.gz.md5	51 B	2019/4/2 下午7:28:00
 env_nt.06.tar.gz	1.1 GB	2019/4/2 下午7:29:00
 env_nt.06.tar.gz.md5	51 B	2019/4/2 下午7:29:00
 env_nt.07.tar.gz	1.1 GB	2019/4/2 下午7:31:00
 env_nt.07.tar.gz.md5	51 B	2019/4/2 下午7:31:00

单机版的Blast使用（四）

核酸序列：

```
$ ./formatdb -i sequence.fa -p F -o T/F -n  
db_name
```

蛋白序列：

```
$ ./formatdb -i sequence.fa -p T -o T/F -n  
db_name
```

单机版的Blast使用（五）

4. 执行Blast比对

获得了单机版的Blast程序，解压开以后，如果有了相应的数据库（db），那么就可以开始执行Blast分析了。

单机版的Blast程序包，把基本的blast分析，包括blastn，blastp，blastx等都整合到了blastall一个程序里面。

单机版的Blast使用（六）

以下是一个典型的blastn分析命令：

(待分析序列seq.fa，数据库nt_db)

```
$/blastall -p blastn -i seq.fa -d nt_db -w 7 -e 10 -o
```

程序名	输入	数据库	窗口	e值	输出
-----	----	-----	----	----	----

seq.blastn.out

该命令的意思是，对seq.fa文件中的核酸序列对nt_db数据库执行blastn搜索，窗口大小是7，e值限制是10，输出的结果保存到文件seq.blastn.out 中。

单机版的Blast使用（七）

5.Blastall的常用参数

- p 程序名应该是blastn, blastp, blastx, tblastn, tblastx中的一个
- d 数据库名称，默认nr
- i 查询序列文件，默认stdin
- e E值限制，默认10
- o 结果输出文件，默认stdout
- F 过滤选项，默认T
- a 选择进行运算的CPU个数

Blast+

← → ↻ ⓘ 不安全 | ftp://ftp.ncbi.nlm.nih.gov/blast/executables/blast+/2.2.31/

/blast/executables/blast+/2.2.31/ 的索引

 [上级目录]

名称	大小	修改日期
 ChangeLog	85 B	2015/6/15 上午8:00:00
 README-win32.txt	682 B	2015/9/10 上午8:00:00
 ncbi-blast-2.2.31+-2.src.rpm	14.4 MB	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-2.src.rpm.md5	63 B	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-2.x86_64.rpm	167 MB	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-2.x86_64.rpm.md5	66 B	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-src.tar.gz	18.1 MB	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-src.tar.gz.md5	64 B	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-src.zip	21.3 MB	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-src.zip.md5	61 B	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-universal-macosx.tar.gz	244 MB	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-universal-macosx.tar.gz.md5	77 B	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-win64.exe	68.9 MB	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-win64.exe.md5	63 B	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-x64-linux.tar.gz	167 MB	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-x64-linux.tar.gz.md5	70 B	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-x64-win64.tar.gz	68.7 MB	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+-x64-win64.tar.gz.md5	70 B	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+.dmg	246 MB	2015/6/15 上午8:00:00
 ncbi-blast-2.2.31+.dmg.md5	57 B	2015/6/15 上午8:00:00

- 任务

- 比对 序列**S** 和 数据库**D**

- 找出 **D**中和**S**最相近的一系列序列

Blast+构建数据库

- 核酸

- `Makeblastdb.exe` `-in` `DNAfastaFile.fasta`
`-dbtype` `nucl` `-out` `DNAfastaFileDatabase`

- 蛋白

- `makeblastdb -in PROfastaFile.fasta -dbtype`
`prot -out PROfastaFileDatabase -logfile log.txt`

Blast+ 比对

- **blastn** –query DNAquery.fasta –db DNAfastaFileDatabase –out DNAVsDNADatabase.txt
- **blastp** –query PROquery.fasta –db PROfastaFileDatabase –out PROVsPRODatabase.txt
- **tblastn** –query PROquery.fasta –db DNAfastaFileDatabase –out PROVsDNADatabase.txt